Runlength Histogram Image Signature for Perceptual Retrieval of Architectural Floor Plans

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Abstract. This paper proposes a runlength histogram signature as a perceptual descriptor of architectural plans in a retrieval scenario. The style of an architectural drawing is characterized by the perception of lines, shapes and texture. Such visual stimuli are the basis for defining semantic concepts as space properties, symmetry, density, etc. We propose runlength histograms extracted in vertical, horizontal and diagonal directions as a characterization of line and space properties in floorplans, so it can be roughly associated to a description of walls and room structure. A retrieval application illustrates the performance of the proposed approach, where given a plan as a query, similar ones are obtained from a database. A ground truth based on human observation has been constructed to validate the hypothesis. Additional retrieval results on sketched building's facades are reported qualitatively in this paper. Its good description and its adaptability to two different sketch drawings despite its simplicity shows the interest of the proposed approach and opens a challenging research line in graphics recognition.

Keywords: Graphics Recognition; Graphics Retrieval; Image Classification

1 Introduction

Aesthetics is a branch of philosophy devoted to beauty, scientifically defined as the study of sensory or sensori-emotional values, sometimes called judgements of sentiment and taste. Visual aesthetics is an emerging topic in computer vision [1, 8]. The general idea is to use machine learning techniques to score the aesthetics of images in terms of visual cues as color, shape and texture, given a subjective human characterization. In document analysis, and in particular in graphics recognition, to the best of our knowledge the concept of aesthetics has not been developed. Some promising approaches have been proposed on perceptual document analysis to segment text lines [7], or to interpret sketches [9]. In these works the structure of the document is analyzed in terms of salient

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Abstract. 本文在检索场景中提出了一种基于游程长度直方图的特征描述子，用于表征建筑平面图的感知特性。建筑图纸的风格主要由线条、形状及纹理的感知所决定，而这些视觉刺激构成了诸如空间属性、对称性、密度等语义概念的基础。我们采用沿垂直、水平及对角方向提取的游程长度直方图来刻画平面图中的线条与空间特性，从而可粗略地反映墙体与房间结构的布局。通过一个检索应用展示了所提方法的性能：给定一张作为查询的平面图，即可从数据库中检索出与其相似的平面图。为验证该假设，我们构建了一个基于人工观察的真值数据集。此外，本文还定性报告了针对建筑立面草图的额外检索结果。尽管方法简单，但其良好的描述能力以及对两类不同草图的适应性，凸显了该方法的研究价值，并为图形识别领域开辟了一条富有挑战性的研究方向。

Keywords: 图形识别；图形检索；图像分类

1 Introduction

美学是哲学的一个分支，其科学定义为对感性或感性—情感价值的研究，有时也被称为情感与品味的判断。视觉美学是计算机视觉领域中一个新兴的研究课题[1, 8]。其基本思想是，在给定主观的人类评价标准的前提下，利用机器学习技术，基于颜色、形状和纹理等视觉线索，对图像的美学质量进行评分。在文档分析领域，尤其是在图形识别方面，据我们所知，美学这一概念尚未得到系统性的研究与拓展。尽管如此，已有若干颇具前景的方法被提出，用于感知型文档分析，例如文本行的分割[7]，或草图的语义理解[9]。在这些研究中，文档的结构是通过显著性特征来进行分析的。

objects, following perceptual grouping rules. We propose the use of perceptual organization, i.e. concepts like saliency, closure, repetition, alignment of geometric primitives, etc. as the basis of aesthetics in graphical documents.

Architecture is a visual art where a creative process results in a design that encompasses aesthetics and function. Hence, a composition is influenced by the standards of beauty that can vary depending on the social, cultural and temporal context, but also by the function of the building and its parts. In the conceptualization stage, architects use CAD software or just sketching interfaces for designing new constructions, projecting it in different views, namely the floor plan or the facade. In this stage, a blank paper is considered as a canvas, and the line drawings are like a painting. The language of architecture consists of basic visual tokens as lines, shapes and texture. Lines can be thick or thin, straight or curved, jagged or smooth, dotted or continuous. Shapes define symbols (structure, furniture, utilities). Texture describes materials and object surfaces. The perception of the spatial arrangement and combination of the basic visual elements gives rise to concepts as symmetry, balance, rhythm, proportion, space, etc. A style is characterized by these concepts, influenced by cultural, societal and temporal trends. Roughly speaking, gothic constructions have geometrically ordered and dense ornaments and vertical emphasis; American colonial house plans feature symmetry in doors and windows; Mediterranean style plans include big spaces like patios or courtyards.

This paper is an early attempt to introduce visual aesthetics in graphics recognition, in particular in line drawings of architectural plans in a retrieval scenario. In this context, aesthetics can be also seen as a drawing style of the architect. Our hypothesis is that an architect can cast queries in large databases in terms of a sample document and retrieve the architectural drawings from the database having a similar style. The state of the art on floor plan retrieval consists in formulating queries in terms of the functional view of the building. Hence, using ontology models, the user can search for similar designs in terms of the number of rooms, the building symbols, etc. [11]. The approach presented in this paper is appearance-based, so we don't recognize building elements, but correlate perceptual semantic concepts with features extracted from raw images. We focus on two topics, namely spaces and lines. Space refers to the size, the layout, or the shape of rooms. Lines in a broader sense give information of the walls width, texture, or length. Perceptually, spaces and lines give an idea of the building structure (e.g. big squared rooms with thick external walls could be stated as a query in terms of spaces and lines). The structure can be associated to the function of the building (a public theater will have big spaces, a bungalow small ones and thin lines defining the walls).

We propose a very simple image signature model, but descriptive enough to represent semantic topics related to some space and line properties. We propose a runlength histogram descriptor. The intuitive idea is that thick lines representing orthogonal walls generate a high frequency of the codeword of long runs in vertical and horizontal direction, and another high frequency in the codeword of runlenghts corresponding to the width of the walls. On the other hand, big rooms

对象，并遵循知觉分组规则。我们提出将知觉组织，即显著性、闭合性、重复性、几何基元的对齐等概念，作为图形文档中美学的基础。

建筑是一门视觉艺术，其创作过程所形成的方案兼具美学与功能。因此，构图既受到因社会、文化及时代背景而异的审美标准的影响，也受到建筑及其各组成部分功能的制约。在概念化阶段，建筑师通常借助CAD软件或手绘界面来设计新建筑，并从不同视角进行表达，如平面图和立面图。在此阶段，白纸被视为画布，线条绘制则如同绘画一般。建筑语言由基本的视觉要素构成，包括线条、形状和质感。线条可粗可细、或直或曲、或锯齿状或平滑、或点状或连续；形状用于界定各类符号（如结构、家具、设备等）；质感则用来描述材料及物体表面。对空间布局以及这些基本视觉要素组合方式的感知，会衍生出诸如对称、平衡、节奏、比例、空间等概念。一种风格正是由这些概念所界定，并深受文化、社会及时代潮流的影响。粗略而言，哥特式建筑以几何有序且繁密的装饰及垂直向的强调为特征；美式殖民风格的住宅平面布局在门窗布置上讲究对称；地中海风格的平面则常包含庭院或天井等大尺度空间。

本文是将视觉美学引入图形识别领域的一次早期尝试，尤其针对检索场景下的建筑平面线图。在这一背景下，美学亦可被视为建筑师的一种绘图风格。我们的假设是：建筑师能够以某份样本文档为基准，在大型数据库中构建查询，并检索出风格相似的建筑图纸。当前楼层平面检索领域的研究现状主要基于建筑的功能视图来构建查询；因此，借助本体模型，用户可按房间数量、建筑符号等特征搜索相似设计[11]。本文所提出的方案则以外观为基础，不直接识别建筑构件，而是将感知语义概念与从原始图像中提取的特征进行关联。我们重点关注两个方面：空间与线条。其中，“空间”指房间的尺寸、布局或形状；“线条”则广义上反映了墙体的宽度、纹理或长度。从感知层面看，空间与线条共同传达了建筑的结构信息（例如，大而方正的房间搭配厚重的外墙，便可作为基于空间与线条的查询条件）。这种结构又可与建筑的功能相联系——公共剧院通常具有较大的空间，而平房则空间较小且墙体线条纤细。

我们提出了一种非常简单的图像特征表示模型，但其描述能力足以表征与空间及线条属性相关的语义主题。我们引入了一种游程长度直方图描述子。其直观思路是：表示正交墙体的粗线条会在垂直和水平方向上产生长游程码字的高频分布，同时在对应墙体宽度的游程长度码字上也呈现另一高频分布。另一方面，大房间

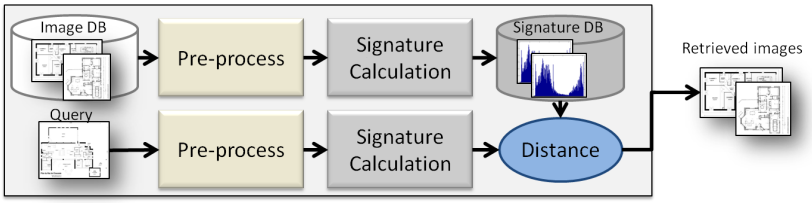


Fig. 1: Pipeline of the method

(spaces) are represented by histogram maxima in the runlengths corresponding to their size. The retrieval model proposed is a query by example framework. Given a query architectural image, we retrieve images that are similar in terms of the space and line topics.

This model has also been tested for retrieving similar sketch drawings of building facades. Qualitative results on arbitrarily downloaded images from the Internet are reported in the paper. Our signature is able to describe in a similar way not only buildings of a similar architectural style, but also in terms of the drawing technique.

The rest of the paper is organized as follows. In Sec. 2 we first review the runlength histogram descriptor and afterwards the approach for retrieving architectural drawings is described. In Sec. 3 the performance evaluation protocol and the experiments are presented. Finally Sec. 4 draws the conclusions and outlines the future work.

2 The runlenght histogram representation

A runglenth is the number of linewise consecutive pixels with the same value in a given direction. Given a direction d that defines the scanpath, the *runlength histogram*, denoted as H_d, encodes the length frequencies of the runs in the image extracted linewise at direction d. Thus, $H_d[i]$ counts the number of runs of length i at direction d. Runs can be extracted from foreground or background images. Thus, separate histograms can be considered, or they can be accumulated in a single one depending on the application. To avoid confusions, we will denote as H_d^f, H_d^b or H_d the foreground, background or joint histograms respectively.

Runlength analysis has been used in document analysis with different purposes. Due to the simplicity in the computation, vertical and horizontal runs are the most usually used. In [5], runlength histograms are used to classify between textual and non-textual blocks in administrative documents. Lately, in [10], runlength projections are used to detect page frames in double-page scanned documents. More recently, in [2], the authors suggest the use of runlengths in the detection of potential wall-elements in architectural floor plans. Finally, in [4], runlength histograms are introduced as whole page document description. Here, foreground runlengths at different scales using spatial pyramids are concatenated and used as signature for efficient classification and retrieval tasks in

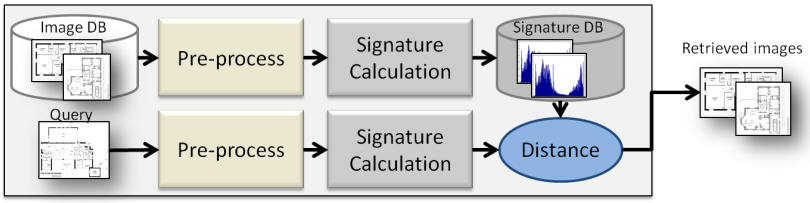


图1：方法流程图

空格通过与其尺寸相对应的游程长度直方图峰值来表示。所提出的检索模型是一种基于示例的查询框架。给定一张查询建筑图像，我们检索在空间与线条主题上与其相似的图像。

该模型还经过了用于检索建筑立面相似草图的测试。论文中报告了基于互联网上任意下载图像的定性实验结果。我们的特征描述不仅能够刻画具有相似建筑风格的建筑物，还能反映其绘图技法的共性。

本文的其余部分安排如下：在第2节中，我们首先回顾了游程长度直方图描述子，随后介绍了用于检索建筑图纸的方法；在第3节中，给出了性能评估方案并展示了实验结果；最后，第4节总结全文并展望未来工作。

2 The runlenght histogram representation

游程长度是指在给定方向上具有相同值的连续像素的行数。给定一个用于定义扫描路径的方向矢量d，记为$runlengthhistogram$的H_d编码了沿方向d逐行提取的图像中各游程的长度频次。因此，$H_d[i]$统计了在方向d上长度为i的游程数目。游程既可从前景图像中提取，也可从背景图像中提取。据此，可根据具体应用分别考虑构建独立的直方图，或将其合并为单一直方图。为避免混淆，我们将分别用H_d^f、H_d^b和H_d表示前景直方图、背景直方图以及联合直方图。

游程分析已在文档分析中被用于多种目的。由于计算简便，垂直游程和水平游程是最为常用的。在文献[5]中，游程直方图被用于区分行政文档中的文本块与非文本块。近期，在文献[10]中，游程投影被用于检测双页扫描文档中的页面边框。更近一些，在文献[2]中，作者提出将游程用于建筑平面图中潜在墙体元素的识别。最后，在文献[4]中，游程直方图被作为一种全页文档的描述方法引入；在此，通过空间金字塔在不同尺度下提取的前景游程被拼接在一起，并用作高效分类与检索任务的特征表示。

large-scale collections. Strongly based in this idea, we use runlength projections as a perceptual signature model for architectural floor plans.

The pipeline of the process, shown in Fig. 1, can be seen as a specific case of existing CBIR strategies. Firstly, all the images are preprocessed for ease of computation. Then, for every floor plan, its signature is calculated by generating the runlength histogram. Given a query example, the method returns a ranked list of the most similar images. Let us further describe the steps of the process.

Preprocess

Floor plan images are first binarized using the Otsu method to be able to extract runs. Afterwards, text-graphic separation is performed to filter out text components, because do not appear in all the images and can slightly bias their global perception. In addition to that, images are resized. Thus, the height of the images has been fixed to 1200 pixels meanwhile the width is rescaled dynamically to keep the aspect ratio. With this, we preserve the same image proportions as the ones used to generate the ground truth, in which the images were shown to the observers in a fixed screen resolution of 1900×1200. The details of the ground truth generation are explained in Sec. 3.1.

Signature model

The signature model to describe every image is extracted as it is shown in Fig. 2. A histogram of runs is calculated in the horizontal, vertical and both diagonals for both, foreground and background layers. Histograms are quantized in bins, experimentally determined, distributed in a logarithmic scale as follows:

$$[1], [2], [3-4], [5-8], [9-16], ..., [257, -].$$

Then, the signature S of an image j is the concatenation of its directional histograms ordered as follows:

$$S_j = [H^f_{0^\circ} : H^f_{45^\circ} : H^f_{90^\circ} : H^f_{135^\circ} : H^b_{0^\circ} : H^b_{45^\circ} : H^b_{90^\circ} : H^b_{135^\circ}]$$

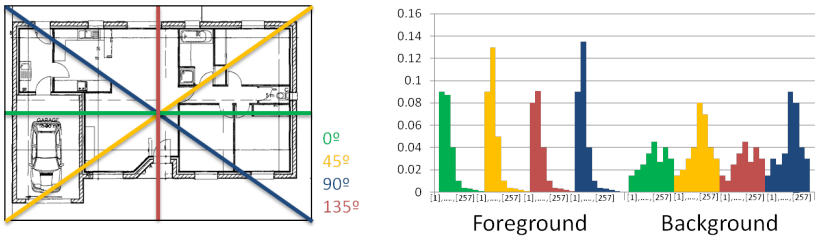


Fig. 2: Signature extraction

大规模数据集。基于这一理念，我们采用游程长度投影作为建筑平面图的感知特征模型。

如图1所示，该流程的管道可被视为现有CBIR策略的一个特例。首先，对所有图像进行预处理以简化计算；随后，针对每一张平面图，通过生成游程长度直方图来计算其特征签名。给定一个查询样本，该方法将返回一张按相似度排序的图像列表。下面将进一步详细描述该流程的各个步骤。

Preprocess

首先，采用Otsu方法对平面布局图像进行二值化处理，以便提取连续的像素行。随后，进行文本与图形的分离，以滤除文本成分，因为文本并非在所有图像中均出现，并且可能对其全局感知产生轻微偏差。此外，还对图像进行了尺寸调整：将图像高度固定为1200像素，同时动态缩放宽度以保持宽高比不变。通过这一处理，我们确保了图像的比例与生成真值时所使用的比例一致；在生成真值的过程中，这些图像以1900×1200的固定屏幕分辨率呈现给观察者。真值生成的具体细节详见第3.1节。

Signature model

如图2所示，提取用于描述每幅图像的特征向量。分别对前景层和背景层，在水平方向、垂直方向以及两条对角线上计算游程直方图。直方图按实验确定的区间进行量化，并以对数尺度分布如下：

$$[1], [2], [3-4], [5-8], [9-16], ..., [257, -].$$

随后，图像 j 的特征向量 S 是由其方向直方图按如下顺序拼接而成：

$$S_j = [H^f_{0^\circ} : H^f_{45^\circ} : H^f_{90^\circ} : H^f_{135^\circ} : H^b_{0^\circ} : H^b_{45^\circ} : H^b_{90^\circ} : H^b_{135^\circ}]$$

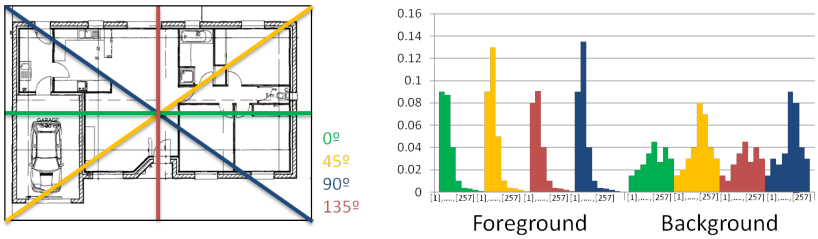


图2：特征提取

Each signature contains 2×4×10 = 80 dimensions. Notice that differently than [4], no spatial information is included in the histogram since there is a lack of correlation between the building structure and its location within the floor plan; as common office-documents do (logos and titles at the top, signatures at the bottom, etc). In addition to that, foreground and background histograms are L1 normalized separately. Our intention is to equilibrate the relevance of the information conveyed by the background and foreground runs, with much higher frequencies in the background ones.

Retrieval

Let P and Q be the signatures of two different floor plan images. Their similarity is calculated by means of the χ^2 distance:

$$\chi^2(P,Q) = \frac{1}{2}\sum_i \frac{(P_i - Q_i)^2}{(P_i + Q_i)}. \quad (1)$$

Given a query instance, the system ranks the rest of the images in the collection according to their affinity to the query, being the first the most similar and the last the most dissimilar.

3 Experiments

3.1 Formulation of the experimental framework

The starting hypothesis of this work is that similarity between architectural plans can be formulated in terms of perceptually dominant visual cues, without interpreting building elements. The runlength histogram descriptor proposed in this paper is a computational model for the perceptual concepts of space and lines. In a retrieval scenario, given a query floorplan image, the proposed method ranks the database images in terms of the runlength histogram similarity. To validate this output in terms of the visual perception, a ground truth based on subjective human assessment was required. This ground truth was constructed with the participation of human observers that classified images in terms of visual aesthetics. Although it is a subjective assessment, we aimed at statistically corroborate that our computational model validates the hypothesis.

A key issue of the ground truth creation is the design of the procedure to collect the user classification of images. A number of considerations must be made. First, since the goal of the users is to classify images with perceptual features and without interpreting the building elements of the architectural drawings, a pre-attentive experiment was conducted. In cognitive vision, pre-attentive processing is the unconscious accumulation of visual information. To force the users to classify images in a pre-attentive way, they have to be displayed in a short lapse of time (no more than a second). A second consideration in the procedure is the way how images are shown to users. Pairwise learning is usually used in the literature [3]. It consists in decomposing a multi-class classification problem

每个签名包含2×4×10 = 80维特征。需要注意的是，与文献[4]不同，本方法的直方图中未包含空间信息，因为建筑结构与其在平面布局中的位置之间缺乏相关性；而常见的办公文档则存在此类相关性（如页眉处的标志和标题、页脚处的签名等）。此外，前景直方图与背景直方图分别进行L1归一化处理。我们的目的是均衡背景区域与前景区域所携带信息的重要性，其中背景区域的出现频率远高于前景区域。

Retrieval

设P和Q分别为两张不同楼层平面图的特征向量。它们之间的相似性通过χ^2距离来计算：

$$\chi^2(P,Q) = \frac{1}{2}\sum_i \frac{(P_i - Q_i)^2}{(P_i + Q_i)}. \tag{1}$$

给定一个查询实例，系统会根据其余图像与该查询的相似度对其在数据集中进行排序，其中排在首位的图像最为相似，排在末位的图像则最为相异。

3 Experiments

3.1 Formulation of the experimental framework

本研究的初始假设是，建筑平面图之间的相似性可以在不解读建筑构件的情况下，通过感知上占主导地位的视觉线索来表征。本文提出的游程长度直方图描述子是一种用于刻画空间与线条等感知概念的计算模型。在检索场景中，给定一张查询楼层平面图，所提出的方法依据游程长度直方图的相似度对数据库中的图像进行排序。为从视觉感知的角度验证该输出的有效性，需要构建一个基于主观人类评估的真值数据集。该真值数据集是在人类观察者的参与下生成的，他们根据视觉美学对图像进行了分类。尽管这是一种主观评估，但我们旨在通过统计分析证实，我们的计算模型能够支持上述假设。

构建真实标签的一个关键问题在于设计用于收集用户对图像分类的流程。在此过程中需要考虑诸多因素。首先，由于用户的任务是基于感知特征对图像进行分类，而不涉及对建筑图纸中各构成要素的解读，因此我们开展了一项前注意实验。在认知视觉领域，前注意加工是指视觉信息的无意识累积过程。为促使用户以前注意的方式对图像进行分类，需在极短的时间内（不超过一秒钟）呈现图像。该流程的另一项考量是图像向用户呈现的方式。文献中通常采用成对学习方法[3]，其核心在于将多分类问题分解为……

into a set of pairwise problems. Therefore given a query, instead of ranking or classifying the whole set of images it is simpler to compare alternatives in a pairwise way. Finally, a third consideration is the question that is made to the users. They can just be asked to assess if a pair of images are similar or not, or a more focused question that affects the observation of the observer (e.g. "do you consider that images are similar in terms of space distribution?", or "don't consider the external shape of the plan when assessing the similarity between images").

Taking into account the above considerations, our ground truth creation was conducted as follows. 20 users participated in the experiment. A database of 39 floor plan images drawn by different architects was used. 6 query images were selected. Each run (different user) showed 234 pairs of images. One of the images was always a query image. Each pair of images was displayed during a second. The user was requested to label each pair of images regarding whether they were very similar, fairly similar or completely dissimilar. At the end, the different labels given by the users to the images where combined averaging the labels so a ranking was generated for each query image.

The human observers selected for generating the ground truth were people aged between 25 and 35 (5 of them were PhD, 14 were PhD students and one had College Studies). None of the volunteers had studied the degree in Architecture and Urban Planning or had any knowledge related with floor plans.

We have formulated two different questions to the users. The question **A**, formulated to 10 observers, was: *"Do you consider that these floor plans are from the same architect or architecture studio?"*. The objective of this question was to cluster the floor plans according to the different architectural styles. On the other hand, to the rest of the observers we asked the question **B**: *"Do you consider that these floor plans are similar?"*. Inversely from question **A**, here the aim is to give more freedom to the observers when considering the meaning of *similar*. Nevertheless, in order to narrow the limits, we suggested to disregard the exterior shape of the building and the size of the plan.

Both formulated questions allowed us to generate three different ground truth rankings. The *gtA*, taking into consideration the answers from the observers asked with question A. The *gtB*, considering the answers to question B. And *gtAB* considering the answers from all the observers that have participated in the experiment, independently of the question formulated to them.

3.2 Performance evaluation protocol

Once the ground truth has been collected after the different user observations as described in the previous section, let us describe how the performance of the proposed approach has been measured. A retrieval process consists in sorting the images of the database according to the similarity to the query. The upper is an image in the ranked list, the more similar is to the query. The ground truth has collected subjective assessments of similarity between the six queries and the database images. These scores given by the users can be turned into ranked

将其转化为一系列成对问题。因此，给定一个查询时，与其对整个图像集合进行排序或分类，不如以成对比较的方式更为简便。最后，第三个需要考虑的因素是向用户提出的问题类型。可以仅要求用户判断一对图像是否相似，也可以提出更具针对性的问题，从而影响观察者的判断（例如：“您认为这两幅图像在空间分布上是否相似？”或“在评估图像相似性时，请勿考虑平面的外部形状”）。

综合上述考虑，我们的真值构建过程如下进行：共有20名用户参与了实验。实验采用了一套由不同建筑师绘制的包含39张平面图的图像数据库，并从中选取了6张查询图像。在每次试验（针对不同用户）中，共呈现234对图像，其中每对图像均包含一张查询图像；每对图像的显示时长为1秒。要求用户对每对图像进行标注，判断其相似程度为“非常相似”、“较为相似”或“完全不相似”。最终，将各用户对同一图像给出的不同标注结果进行平均处理，从而为每张查询图像生成一个排序结果。

用于生成真实标签的人类观察者年龄介于25至35岁之间（其中5名为博士，14名为博士研究生，另有1名为大学在读生）。所有志愿者均未接受过建筑与城市规划专业的系统学习，且不具备与平面图相关的专业知识。

我们向用户提出了两个不同的问题。针对10位观察者提出的问题**A**为：*”Do you consider that these floor plans are from the same architect or architecture studio?”*。该问题的目的是根据不同的建筑风格对平面布局进行聚类。另一方面，对于其余观察者，我们提出了问题**B**：*”Do you consider that these floor plans are similar?”*。与问题**A**相反，此问题旨在使观察者在理解*similar*时拥有更大的自由度。尽管如此，为了缩小范围，我们建议忽略建筑的外部形态及平面的尺寸。

两种提问方式均使我们能够生成三种不同的真值排序：gtA基于对采用问题A的观察者所作回答的考量；gtB基于对问题B的回答的考量；以及$gtAB$，综合考虑所有参与实验的观察者的回答，而不区分其所被提问的具体问题。

3.2 Performance evaluation protocol

在如前一节所述完成不同用户观测并收集到真实标签后，接下来将说明如何评估所提出方法的性能。检索过程是根据与查询图像的相似度对数据库中的图像进行排序：在排序列表中位置越靠前的图像，与查询图像的相似度越高。真实标签包含了六组查询图像与数据库中各图像之间相似度的主观评价；这些由用户给出的评分可以转化为排序结果。

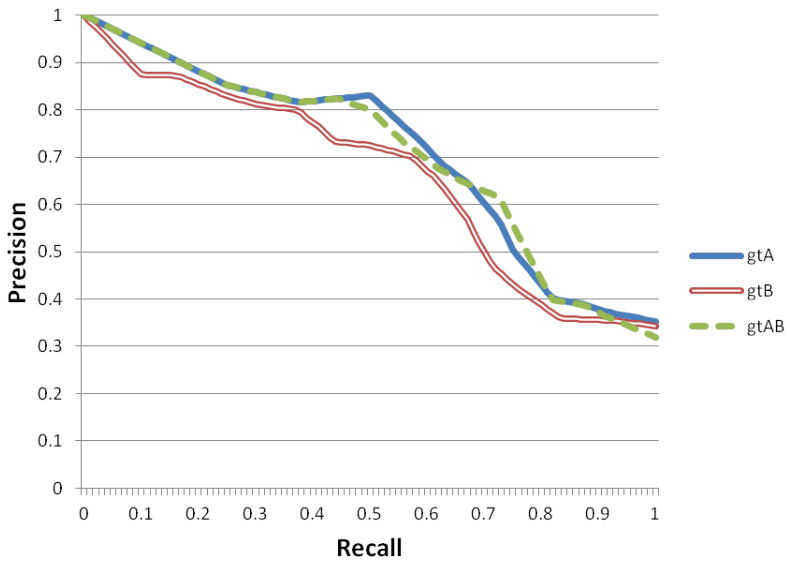


Fig. 3: Precision and recall curves for the three ground truth models.

lists, so given a query, the performance of the approach is measured in terms of ranked list comparison.

A number of distances between rankings have been proposed in the literature [6]. Let S and S' be two ranked lists of items with the particularity of S' being a permutation of S, i.e. both lists contain exactly the same set of items. Let $\sigma(i)$ denote the rank in S' of the element i in S. The *Spearman's footrule* distance between two rankings measures the total element-wise displacement and is given by:

$$F(S, S') = \sum_i |i - \sigma(i)|;$$

This measure only takes into account the order of the elements to assess the similarity between two lists. The order of the elements is given by their similarity score. Hence, if two elements have the same score value, an arbitrary technique has to decide which element is sorted first. This issue can lead to a random-like ranking when several elements are equally scored in a list and then strongly influence in the ranking distance.

This is the case of our ground truth ranking. The affine answers obtained from the observers lead to a ranking with many repeated scores for different images. In order to avoid this problem, we have adapted the distance function between two ranking, considering the same distance value for images with the same score. Thus, let S be a ranking obtained by the system, given a query, and S' its corresponding ground truth, the distance measurement between both

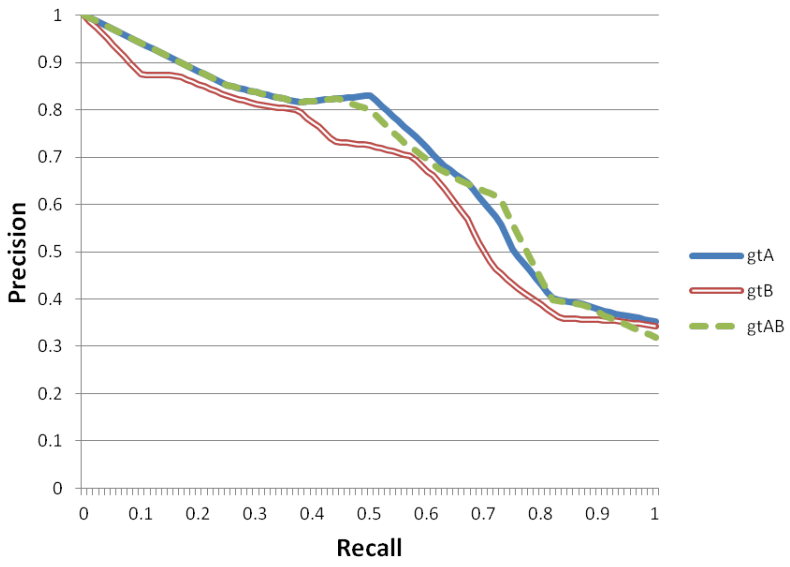


图3 ： 三种真实标签的精确率-召回率曲线 模型。

列表，因此给定一个查询时，该方法的性能通过排序列表的比较来衡量。

文献中已提出了多种排名间距离度量方法[6]。设S和S'为两个项目排序列表，且满足S'是S的一个排列，即这两个列表包含完全相同的项目集合。记$\sigma(i)$为元素i在S中的排名在S'中的位置。两种排名之间的$Spearman's\ footrule$距离度量的是所有元素的逐项位移之和，其表达式为：

$$F(S, S') = \sum_i |i - \sigma(i)|;$$

该度量仅考虑元素的排序来评估两个列表之间的相似性。元素的排序由其相似性得分决定；因此，当两个元素具有相同的得分时，必须采用某种任意规则来确定哪个元素优先排序。这一问题可能导致在列表中存在多个得分相同时出现类似随机的排序，并进而显著影响排序距离的计算结果。

这正是我们的真实排名的情况。观察者给出的仿射评分会导致不同图像出现大量重复的得分，从而影响排序结果。为避免这一问题，我们对两个排序之间的距离函数进行了改进，对于具有相同得分的图像赋予相同的距离值。因此，设S为系统在给定查询时得到的排序，S'为其对应的真实排名，则两者之间的距离度量为：

Table 1: Perceptual ranking results for the six queries.

	gtA	***gtB***	***gtAB***
image1	115	137	154
image2	64	96	94
image3	139	154	154
image4	33	50	53
image5	110	72	99
image6	123	109	108
mean	**97**	**103**	**110**

rankings is given by:

$$F(S, S') = \sum_i min(|i - \hat{\sigma}(i)|),$$

where $\hat{\sigma}(i)$ denotes the class of equivalence in S' of the element i in S, i.e. a representative among the different images in S' having the same score. Formally:

$$\hat{\sigma}(i) = \arg\min_{j \in S'} \Big(i - \sigma(j) | p_{S'}(j) = p_{S'}(\sigma(i)) \Big),$$

where $p_{S'}(j)$ and $p_S(\sigma(i))$ denote the scores of elements j and i of S'.

3.3 Results and Discussion

In Table 1 we show the ranking results obtained for six different queries. The numerical values denote the distance between the system retrieval and the three different ground truth rankings: *gtA*, *gtB* and *gtAB*. In the hypothetical case of a resulting list sorted inversely to the ground truth the score would be 760. When studying the results, the main sorting differences occur in lasts positions of the lists, where most dissimilar images are ranked. Meanwhile, in the first positions, the most similar images are matched for all the queries, a fact that is corroborated by the precision and recall curves extrapolated for all the queries shown in Fig. 3. In addition to that, we also observe that results are slightly different depending on the groundtruth compared with, being *gtA* the one that leads to the best ranking. In retrieval terms, the mean average precision (mAP) obtained also varies depending on the ground truth model. Thus, the mAP scores are 0.67 when compared to *gtA*, 0.63 to *gtB*, and 0.66 to gtAB.

In qualitative terms, we show in Fig. 4 the six queries and their three respectively most similar images retrieved by the system. As it can be seen in every case, the three retrieved images belong to the same collection of the query –share the same drawing style–, a fact that ratifies the fact just mentioned above regarding the matching accuracy in the first positions of the ranking.

Analysing the experiment performed, we realized about the impact on the system performance of the question formulated to the observers in the ground

表1：六项查询的感知排序结果。

	gtA	***gtB***	***gtAB***
image1	115	137	154
image2	64	96	94
image3	139	154	154
image4	33	50	53
image5	110	72	99
image6	123	109	108
mean	**97**	**103**	**110**

排名由下式给出：

$$F(S, S') = \sum_i min(|i - \hat{\sigma}(i)|),$$

其中，$\hat{\sigma}(i)$ 表示元素 i 在 S 中的等价类，即在 S' 中具有相同得分的不同像元中的一个代表。形式化地表述如下：

$$\hat{\sigma}(i) = \arg\min_{j \in S'} \Big(i - \sigma(j) | p_{S'}(j) = p_{S'}(\sigma(i)) \Big),$$

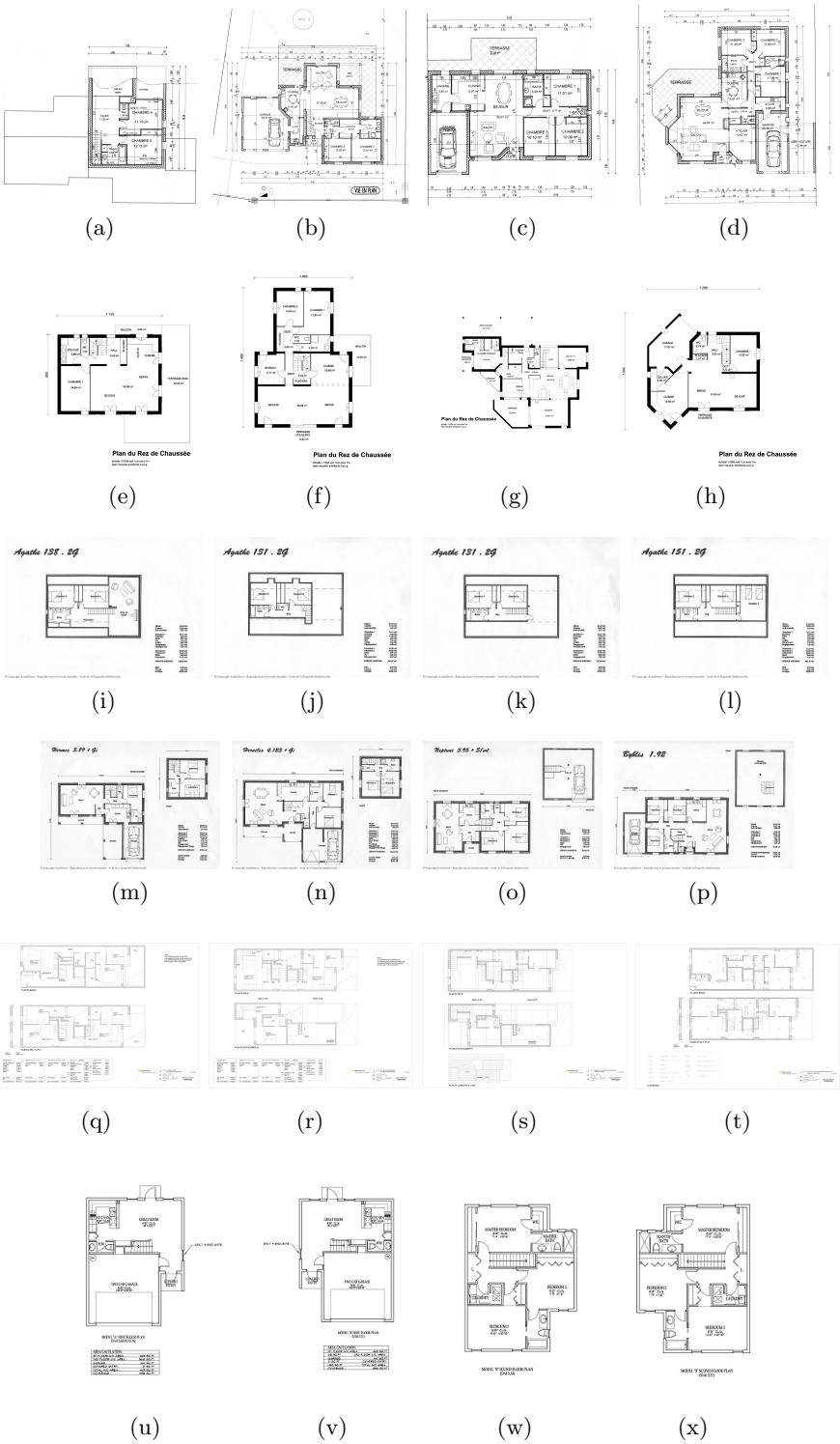
其中，$p_{S'}(j)$ 和 $p_S(\sigma(i))$ 分别表示 S' 中元素 j 和 i 的得分。

3.3 Results and Discussion

在表1中，我们展示了针对六个不同查询所获得的排序结果。数值表示系统检索结果与三种不同的真值排序——gtA、gtB和$gtAB$——之间的距离。在假设情况下，若最终排序列表与真值排序完全相反，则得分将为760。分析结果可知，主要的排序差异出现在列表的末尾位置，即那些最不相似的图像被排列于此；而在列表的前部，对于所有查询而言，最相似的图像均得到了匹配，这一结论也得到了图3中针对所有查询外推得到的精确率—召回率曲线的验证。此外，我们还观察到，不同真值参照下的结果存在细微差异，其中以gtA作为参照时所得的排序效果最佳。从检索性能来看，所获得的平均精度均值（mAP）同样因真值模型的不同而有所变化：与gtA比较时，mAP为0.67；与gtB比较时，mAP为0.63；与gtAB比较时，mAP为0.66。

在定性分析方面，我们在图4中展示了六个查询及其由系统分别检索出的三张最相似图像。如图所示，在每种情况下，这三张检索到的图像均属于与查询相同的图像集——即具有相同的绘图风格——这一事实进一步证实了前述关于排序前几位匹配准确性的结论。

通过对所开展的实验进行分析，我们认识到向地面观察员提出的问题对系统性能所产生的影响。



(a) (b) (c) (d)

(e) (f) (g) (h)

(i) (j) (k) (l)

(m) (n) (o) (p)

(q) (r) (s) (t)

(u) (v) (w) (x)

Fig. 4: Floor plan retrieval. The 6 different queries are shown in left column: (a), (e), (i), (m), (q), and (u). Their respective 3 most similar images sorted by similarity are shown row-wise.

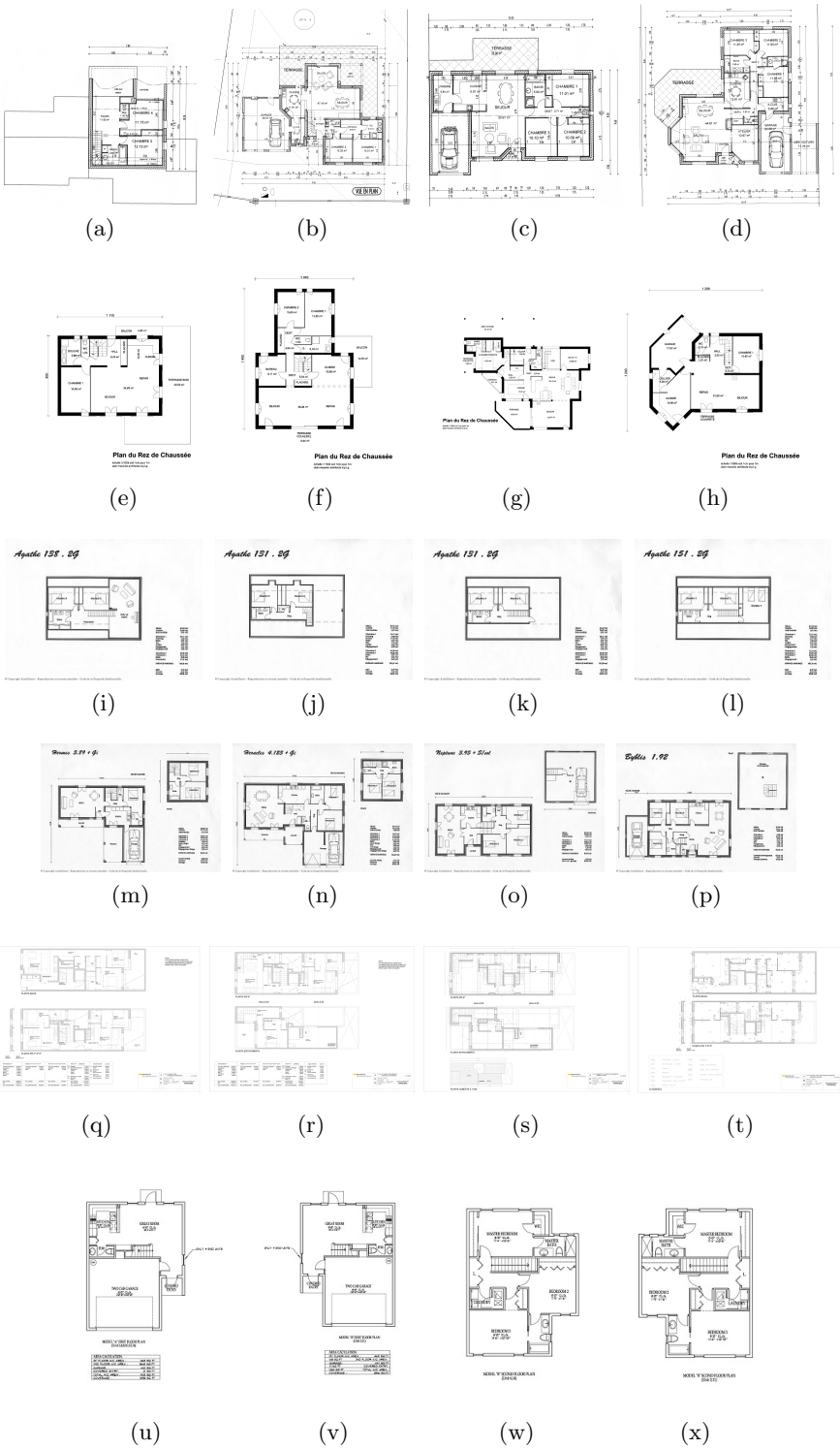


图4：楼层平面图检索。左侧列展示了6个不同的查询：(a)、(e)、(i)、(m)、(q)和(u)。每一行依次展示了按相似度排序的各自3张最相似图像。

truth generation time. It can strongly determine their perception in terms of *similarity*. As it has been shown, different ways to formulate a question aiming for the same answer can imply different interpretations from the observers. Thus, the system performance varies when it is compared with gtA, gtB or $gtAB$. Furthermore, a hight population of observers is needed to generate a trustful ground truth able to smooth the high impact of biased perceptions from certain individuals.

We also used our system for retrieving similar sketch of building's facades given some query images. With this aim, we have downloaded randomly 58 sketches of facades from Google Images [1], which 4 of them are selected arbitrarily to play as queries. The three most similar images to each query are shown in Fig. 5. At a glance, one can note that the retrieved images belong either to similar building styles or drawing technique.

In global terms and despite its simplicity, the proposed document signature is able to fairly model the human perception of similitude in the floor plans framework. Moreover, the qualitative results on facade retrieval shows that it can be extrapolated to additional functionality on other sketched drawings, which opens a new interesting research line in Graphics Recognition.

4 Conclusions

In this paper we have proposed a perceptual model to describe the drawing style in architectural plans. We have considered that semantic concepts associated to lines and space can make two plans similar to the eyes of a human observer in a pre-attentive process. A simple descriptor based in runlength histograms has been proposed. This signature has the ability to roughly capture the properties of the lines (mainly walls) and the spaces (room aspect) of a floorplan design. In a retrieval framework, this descriptor allows to search for perceptually similar plans into a database. The main contributions of the paper have been: first a pioneer approach of graphical document retrieval based on perceptual cues, without interpreting the building elements, i.e. symbols, of the drawing; and second the design of an experimental setup inspired in the pre-attentive theories of cognitive vision.

Although the work is in a preliminary stage, the obtained results are promising. We have only focused on a few number of perceptual features. A complete system that wants to model the style of an architectural drawing combining function and aesthetics should consider other features representing density, proportion, symmetry, etc. The results presented in Sec. 3 allow to conclude that runlength histograms appear as a simple and promising descriptors to capture the intended concepts related to lines and space. When floor plans present orthogonal walls, runlengths characterizing the width or filling texture of walls present high frequencies. On the other hand, big rooms correspond to long high frequent runlengths, and small rooms correspond to mid-length ones.

[1] http://www.google.cat/imghp

真值生成时间。它能够在$similarity$方面强烈影响观察者的认知。如前所示，针对同一答案的不同提问方式可能会导致观察者产生不同的解读。因此，当与gtA、gtB或$gtAB$进行比较时，系统的性能表现会有所差异。此外，为了生成能够有效平抑个别个体偏见认知所带来的显著影响的可信真值，需要足够规模的观察者群体。

我们还利用该系统，根据给定的查询图像检索建筑立面的相似草图。为此，我们从Google图片中随机下载了58幅建筑立面草图[1]，其中任意选取4幅作为查询图像。图5展示了每幅查询图像对应的3幅最相似图像。粗略观察即可发现，检索到的图像要么属于相似的建筑风格，要么采用相近的绘图技法。

从全局角度来看，尽管结构简单，所提出的文档签名方法仍能在平面图框架下较好地模拟人类对相似性的感知。此外，立面检索的定性结果表明，该方法可进一步推广至其他草图绘制领域的功能拓展，从而在图形识别领域开辟一条新的、富有前景的研究方向。

4 Conclusions

本文提出了一种用于描述建筑平面图绘制风格的感知模型。我们认为，与线条和空间相关的语义概念能够在前注意加工过程中使两张平面图在人类观察者眼中显得相似。为此，我们提出了一种基于游程长度直方图的简单描述子，该描述子能够粗略地刻画平面设计中线条（主要指墙体）与空间（房间形态）的特征。在检索框架下，该描述子可用于在数据库中搜索与查询平面图在感知上相似的平面图。本文的主要贡献在于：其一，提出了一种基于感知线索的图形文档检索方法，无需对图纸中的建筑元素（即符号）进行语义解释；其二，设计了一个受认知视觉中前注意理论启发的实验方案。

尽管目前工作尚处于初步阶段，但所获得的结果已显示出良好的前景。我们仅关注了少数几种感知特征；一个旨在同时兼顾功能与美学来建模建筑图纸风格的完整系统，还应考虑其他能够表征密度、比例、对称性等特性的特征。第3节中呈现的结果表明，游程长度直方图作为一种简单而有前景的描述子，能够有效捕捉与线条及空间相关的预期概念。当平面图中墙体呈正交布局时，用于表征墙体宽度或填充纹理的游程长度会出现较高的频率；反之，大房间对应于较长且高频的游程长度，而小房间则对应于中等长度的游程长度。

[1] http://www.google.cat/imghp

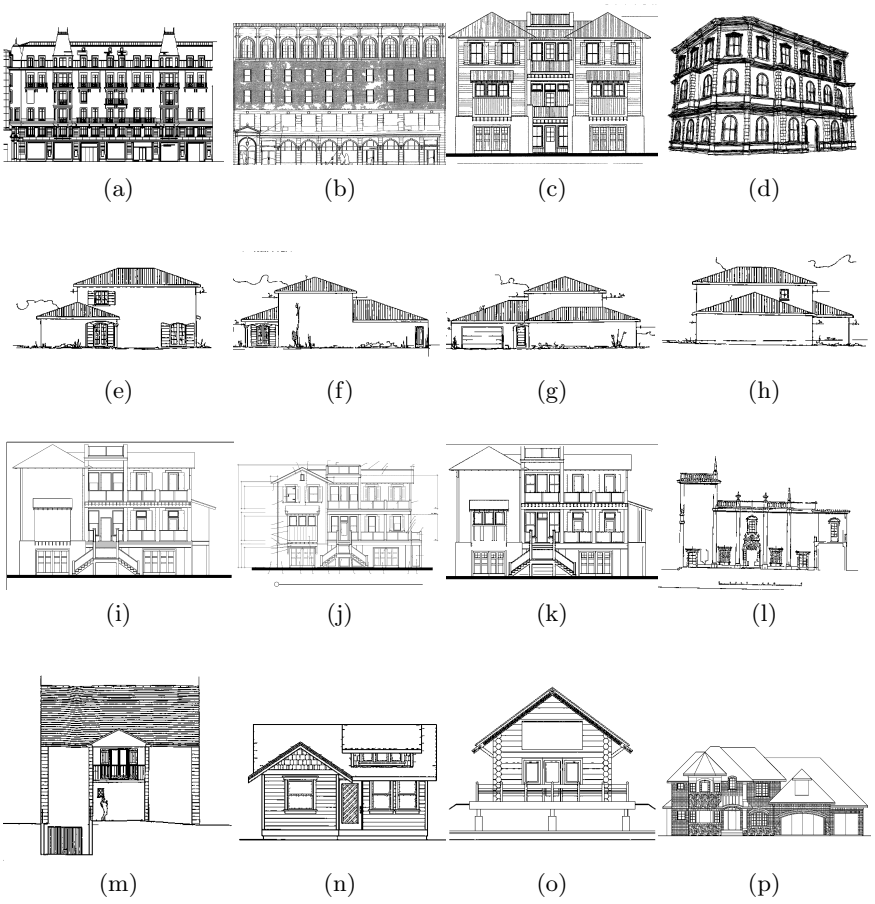


Fig. 5: Retrieval of sketch drawings of building's facades. In (a), (e), (i), and (m) the four different queries are shown. Their respective most similar instances are shown row-wise.

To validate our model, we have collected ground truth based on human observation. A pre-attentive experiment has been conducted, resulting in ranked lists of floor plan images according to the human assessment of their similarity to a given query. We have noticed the subjectivity of this procedure, which corroborates the complexity of human perception in visual aesthetics. We have experimentally observed the relevance of the question made to the users when they are asked to score images.

Additionally, we have shown qualitatively that our model can be extrapolated to other sketch-based drawing retrieval problems. It has been tested successfully to capture similar drawing styles and building types from randomly downloaded facade sketches.

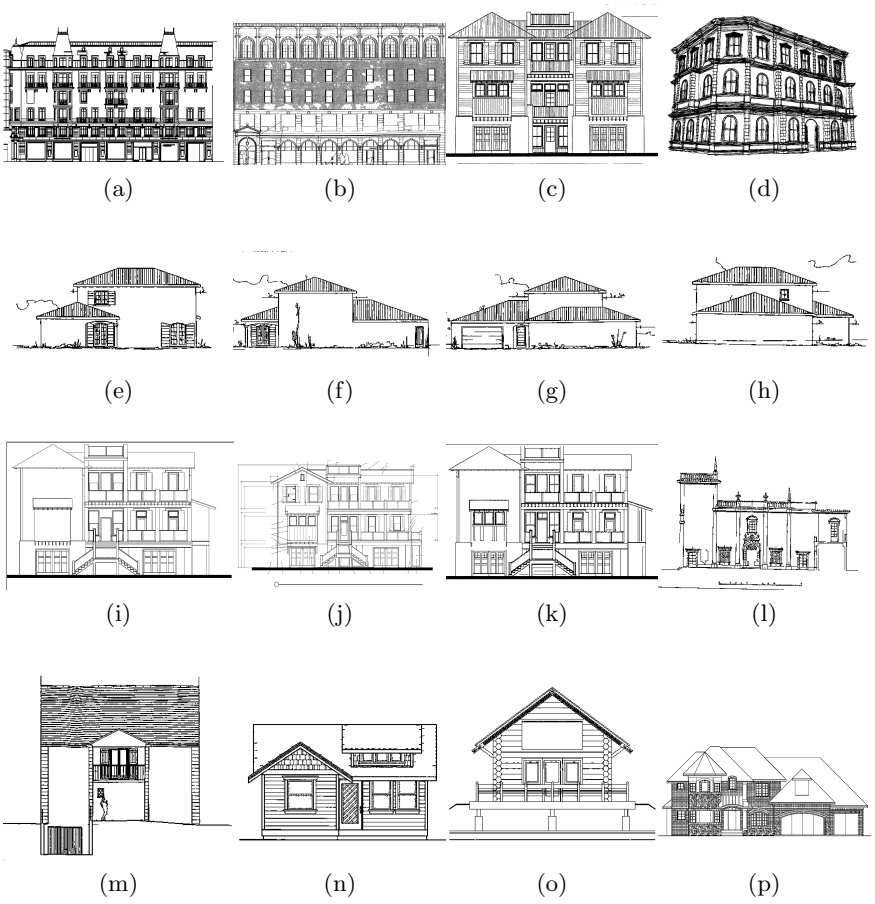


图5：建筑立面草图的检索结果。（a）、（e）、（i）和（m）中分别展示了四个不同的查询，其各自最相似的实例按行展示。

为验证我们的模型，我们基于人工观察收集了真实标签。我们开展了一项前注意实验，根据人类对图像与给定查询相似度的主观评价，生成了平面图图像的排序列表。我们注意到该过程具有一定的主观性，这进一步印证了人类在视觉美学感知方面的复杂性。此外，我们通过实验发现，在要求用户对图像进行评分时，所提问题的表述方式对其评分结果具有显著影响。

此外，我们已从定性层面证明，我们的模型可推广应用于其他基于草图的绘图检索问题。该模型已在随机下载的建筑立面草图中成功捕捉到相似的绘图风格与建筑类型。

As a continuation of the work, we will focus on enhancing our descriptor for architectural drawings. We are considering to describe other semantic concepts to have a more complex model of visual aesthetics characterizing styles of this kind of documents. Although some exchanges have been done with architects, a rigorous validation should be carried on with experts. It is necessary to translate the semantic concepts that guide their perception to visual features, making the proposed model usable in professional tools.

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作为该项工作的延续，我们将重点提升用于建筑图纸的描述符。我们正考虑引入对其他语义概念的刻画，以构建一个更为复杂的视觉美学模型，从而表征此类文档的风格特征。尽管已与建筑师进行了初步交流，但仍需开展严谨的专家验证工作。有必要将指导其感知的语义概念映射为视觉特征，从而使所提出的模型能够在专业工具中得到应用。

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